INCH-POUND
MIL-DTL-24640/12D
22 August 2002
SUPERSEDING
MIL-C-24640/12C
26 June 1995

DETAIL SPECIFICATION SHEET

CABLE, ELECTRIC, 600 VOLTS, LIGHT-WEIGHT, MULTI-PAIR, AWG 26, TYPE 2XO

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24640 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

Construction (Non-Watertight)

First - 19-strand, tin-coated copper conductor, AWG size 26.
Second - Component wires shall be MIL-W-81044/12, except identification shall be in accordance with the codes and methods of MIL-DTL-24640. Testing shall be in accordance with the component wire ratings and requirements listed in this slash sheet. Quality conformance testing shall be in accordance with MIL-W-81044, prior to cabling. Twisted pair identification code, one black and one white, applied using method 6 or telephone color code applied using method 3.
Third - Two conductors twisted together. Pair lay length shall be in accordance with Tables II and III.
Fourth - Cable assembly shall be in accordance with Table III. Cabling sequence shall be consecutive, starting with number 1 from center outward. Fillers may be used as necessary, to form a round cable.
Fifth - Binder tape applied helically with overlap.
Sixth - Braided shield, tin coated copper strands. Braid must be constructed to meet Surface Transfer Impedance requirements as listed herein.
Seventh - Binder tape applied helically with overlap, at manufacturer's option.
Eighth - Crosslinked polyolefin jacket. (See Table I for thickness).

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TABLE I. Construction details.

PIN M24640/12	Type number	Number of component	Diameter over shield (inch) (nom)	Jacket thickness (inch) (min)	Cable diameter (inch) (min)	(max)	Cable weight (pounds per 1000 feet) (max)
-01UO	2XO-6	6	0.220	0.038	0.305	0.329	81.6
-02UO	2XO-18	18	.356	.041	.417	.449	140.0
-03UO	2XO-24	24	.420	.044	.473	.509	175.0
-04UO	2XO-42	42	.536	.044	.565	.609	275.0
-05UO	2XO-60	60	.622	.047	.641	.691	334.0
-06UO	2XO-77	77	.714	.047	.728	.785	395.0

TABLE II. Pair lay [1/]

Pair no.	Pair lay (inches)	Pair no.	Pair lay (inches)	Pair no.	Pair lay (inches)
1	1.5	27	1.0	53	2.5
2	1.0	28	2.0	54	3.5
3	2.0	29	3.0	55	4.5
4	3.0	30	4.5	56	5.5
5	4.5	31	1.0	57	1.0
6	1.0	32	2.0	58	1.5
7	2.0	33	3.0	59	2.0
8	3.0	34	4.5	60	2.5
9	1.5	35	1.0	61	3.0
10	2.5	36	2.0	62	4.5
11	5.5	37	3.0	63	5.5
12	1.5	38	4.5	64	1.0
13	2.5	39	1.0	65	1.5
14	3.7	40	2.0	66	2.0
15	5.5	41	3.0	67	2.5
16	1.5	42	4.5	68	3.0
17	2.5	43	1.0	69	4.5
18	3.7	44	1.5	70	5.5
19	1.5	45	2.0	71	1.0
20	2.5	46	2.5	72	1.5
21	3.7	47	3.0	73	2.0
22	5.5	48	4.5	74	2.5
23	1.0	49	5.5	75	3.0
24	2.0	50	1.0	76	4.5
25	3.0	51	1.5	77	5.5
26	4.5	52	2.0		

[1/] All lays are left-handed.

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TABLE III. Assembly.

Type number	Core	Assembly			
		Layer 1	Layer 2	Layer 3	Layer 4
2XO-6	- - -	Pairs 1-6			
2XO-18	Pair 1	Pairs 2-6	Pairs 7-18		
2XO-24	Pairs 1-2	Pairs 3-10	Pairs 11-24		
2XO-42	- - -	Pairs 1-8	Pairs 9-22	Pairs 23-42	
2XO-60	Pair 1	Pairs 2-7	Pairs 8-19	Pairs 20-36	Pairs 37-60
2XO-77	Pairs 1-3	Pairs 4-12	Pairs 13-27	Pairs 28-49	Pairs 50-77

REQUIREMENTS:

Qualification required.
Construction details: See Tables I, II and III.

Component wire: The component wire shall be per MIL-W-81044 and shall be tested in accordance with the following wire ratings and additional requirements.

Temperature rating: 150°C (302°F) maximum conductor temperature
Voltage rating: 600 volts (rms) at sea level.
Accelerated aging: Oven temperature, 300 ± 2°C (572 ± 3.6°F) for 6 hours.
Blocking: 150 ± 2°C (302 ± 3.6°F)
Flammability: 30 seconds (max); 3.0 inches (max); no flaming of tissue paper.
Humidity resistance: 5000 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Impulse dielectric test:
    Primary insulation (when tested in lieu of spark test):
        6.0 kilovolts (peak) 100 percent test.
    Finished wire: 8.0 kilovolts (peak), 3 kHz spark test
    5.5 kV, 100 percent test.
Insulation resistance: 5000 megohms for 1000 feet (min).
Life cycle: Oven temperature, 200 ± 2°C (392 ± 3.6°F) for 168 hours.
Physical properties of primary insulation:
    Tensile strength, 2500 lb/in^2 (min).
    Elongation, 150 percent (min).
Shrinkage: 0.125 inch maximum at 300 ± 2°C (572 ± 3.6°F).
Spark test primary insulation: 1500 volts (rms), 60 Hz or 3 kHz, 100 percent test.
Surface resistance: 500 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150 ± 2°C (302 ± 3.6°F).
    Maximum change in measurement: 0.060 inch.
Wet dielectric test: 2500 volts (rms).
Wrap test: Mandrel wrap test required.

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EXAMINATION AND TESTS:

Basic electrical:	Requirements:
Conductor Resistance - ohms/1000 feet at 25°C, maximum	44.24
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to shield	1000
Component Shield to shield	Not applicable
Insulation resistance - megohms-1000 feet, minimum	2500
Conductor and shield continuity	No failure
Jacket flaws	No failure
Group A:	
Visual and dimensional	No failure
Capacitance:	
Mutual capacitance - at 1 MHz, picofarads/foot(pF/ft), maximum	33
Capacitance unbalance - percent, maximum	15
Characteristic impedance - at 1 MHz, ohms	75 min, 105 max
Watertightness (see Table IX of MIL-DTL-24640 for limits of water leakage)	Not applicable
Weight	No failure
Group B:	
Cold bending	No failure
Drip	Not applicable
Cross link proof test (percent maximum)	
Jacket ...	50
Tear - pounds per inch thickness, minimum	35
Physicals (unaged)	
Jacket	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Attenuation - at 3 MHz, dB/100 feet, maximum	Not applicable
Gas flame (1 hour)	Not applicable
Group C:	
Physicals (aged)	
Jacket	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Permanence of printing (conductor) method 1 only cycles, minimum	25
Permanence of printing (jacket) - cycles, minimum	125
Shield - conformance to material, construction and coverage	No failure
Heat distortion - percent of unaged, maximum	30

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Cable sealant removability ..... Not applicable
Shrinkage ..... No failure
Mutual inductance - dB/1000 feet, maximum ..... Not applicable
Surface transfer impedance
    Milliohms per meter, maximum 700
    EMP response, dB, minimum 60
Pulse response time - ms/500 feet
    Rise time, maximum ..... Not applicable
    Decay time, maximum ..... Not applicable

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Flame propagation ..... No failure
Aging and compatibility (cable) ..... No failure
Acid gas equivalent - percent, maximum
    Jacket 2
    Fillers 2
    Insulation 18
Halogen content - percent, maximum
    Jacket 0.2
    Fillers 0.2
Hot oil immersion
    Tensile strength - percent of unaged, minimum 50
    Elongation - percent of unaged, minimum 50
Smoke index, maximum
    Jacket 25
    Fillers 45
    Insulation 10
Toxicity index, maximum
    Jacket 5
    Fillers 5
    Insulation 1.5
Durometer (jacket) - (type A) hardness, minimum 80
Weathering (jacket) ..... No failure
Fungus Resistance ..... No failure
Abrasion resistance ..... No failure
Gas Flame (3 hour) ..... Not applicable
Component wire ..... No failure

Unit ordering length: 6 - 24 500 feet nominal
42 - 77 250 feet nominal

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Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2307-012)

Review Activities:
Army - AR, AV, CR, CR4
Navy - CG, EC
DLA - IS

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